

Some useful definitions

Trials factors: Given a single GW real event, multiple pipelines can trigger. Similarly for a noise event. To account for this, the actual rate of false triggers, given a publication threshold, should be corrected by:

$$\text{FAR}_{\text{actual}} = k \times \text{FAR}_{\text{threshold}}$$

A conservative choice for k is the total number of pipelines (+1 given by RAVEN):

- 6 + 1 for CBC
- 2 + 1 for Burst

For instance, for CBC triggers, this choice gives an overall expected rate of triggers of:

- 1 SIGNIFICANT event / month
- 10 non-SIGNIFICANT events / day

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Events clustering: since multiple pipelines can trigger, more than one event can be associated with the same astrophysical event —> events that have a temporal proximity of $[-1,+1]$ from each other are clustered in a single **SUPEREVENT**

On GraceDB events are labeled with a G*** name, superevents with an S*** name

P_astro: Quantifies the probability that the event is of astrophysical origin

- Provided only for CBC pipelines
- Depend on our prior knowledge on the astrophysical rate of each class (BBH, NSBH, BNS) as well as on assumptions on the mass distribution
- Divided in $P(\text{BBH})$, $P(\text{NSBH})$, $P(\text{BNS})$, $P(\text{Terrestrial})$ and they sum to 1